Local Search In Approximate Submodular Maximization and Usage in Diverse Feature Selection

M.Tech Project Stage II Report

Submitted in partial fulfillment of the requirements for the degree of

Master of Technology

by

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To My Mother, My Sister and Sayantani

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Prof. Ganesh Ramakrishnan, who introduced me to this domain. I can't thank him enough for his valuable time, suggestions, help and constructive criticism during the entire work. I would also like give special thanks to my fellow M.Tech student, Pankaj Singh, for having long discussions with me on the topic which helped me understand the material better. Furthermore I would also like to acknowledge with much appreciation the staff of CSE department for all their help in the official procedure.

Special thanks to all my friends in and off campus who helped me in numerous occasions I can't even begin to name.

Finally, this work would not have been possible without the constant support and enthusiasm from my mother, my sister and Sayantani. Thanks for being there for me always.

Abstract

Submodular functions (see Lovász, 1983, Narayanan, 2009) form an interesting field of research due to their mathematical elegance and wide range of applicability in numerous areas, including Machine Learning, Economics, Natural Language Processing, to name a few. Submodular functions' connections with both convexity and concavity have made them ideal choice for candidate set functions in discrete optimization problems. In many practical scenarios, however, exact submodularity appears to be a strict condition to observe. Therefore, in Das and Kempe, 2011, the notion of *approximate submodularity* was introduced in the context of feature selection task for linear regression. In order to promote diversity in the selected features for many practical reasons, a number of submodular (monotone or non-monotone) regularizers were used along with an approximate submodular objective function in the feature selection task for linear regression problems in Das et al., 2012. In that work, for the non-monotone case, a rather expensive local search based algorithm based on Feige et al., 2011 is used in addition to a standard forward selection algorithm to show that it has a provable optimality guarantee for such objective function. In the present work, we show that using a linear-time local search similar to Buchbinder et al., 2012, we can also achieve an optimality guarantee for the such objective function and therefore reduce the complexity of the whole algorithm for the same task without compromising much regarding the performance.

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Chapter 1

Introduction

Feature selection is an important task in the machine learning and statistics community. In standard supervised classification or regression problems, we are given a number observations with, usually, multivariate features (independent variables) and a target variable or class as the dependent variable. The learning task comprises of fitting a mathematical model based on this training data set which is then used for prediction of the target variable for unseen data points. The key measure for this task is to usually reduce the error in prediction or penalty for misprediction. But it is often observed that many of such input features do not play any significant role in the mathematical model being learned, either by not being sufficiently correlated with the target variable and therefore being useless or by being highly correlated with another feature and therefore being redundant. In such scenarios, feature selection task can be used as a pre-processing step for the data set, in which we select, based on some measure, the useful features and get rid of the rest before the learning task begins. Therefore, from system's perspective, essentially feature selection pre-processing can dramatically reduce the complexity of the learning problem. It also helps inferring the model manually where understanding the roles of the input features to predict the target in the model is desired.

1.1 Feature selection - overview

Selecting features from a given set based on certain measures is a combinatorial optimization problem and is usually hard. A naive approach towards feature selection can be a cross validation setting, where all possible combinations of features are used to learn models for prediction and the features corresponding to the model with minimal prediction error are selected. However such a system is infeasible due to the exponential nature of this approach. Also, it requires invoking the learner itself in the pre-processing step and therefore defeats the purpose of feature selection in the first place. Usual methods thereby aim for minimizing a "proxy measure" for prediction error. The two most commonly used algorithms for this purpose are *Forward Selection* and *Backward Elimination*. In Forward Selection case, the approach starts with an empty set in hand and it iteratively keeps adding new features to that set minimizing (or maximizing) some measure until the desired number of features is reached (which usually is a user input). Backward Elimination, on the other hard, takes the same approach from the other direction, *i.e.* start with the entire set and iteratively gets rid of the features one by one until desired number of features are left in the set.

1.1.1 Heuristic for feature selection

Finding the proper measure for feature selection algorithms is fundamentally a heuristic and is specific to the mathematical nature of the model since the notion of dependency or correlation depends on that. For example, consider that a support vector machine (SVM) classifier with some string kernel is being used as the model in order to classify e-mails as spam or ham. A feature selection task here can be to induce a set of keywords which play important role in such classification. Therefore, the appropriate measure for this task should be to look for dependency and correlation in the kernel mapped space. Linear regression, on the other hand, is one of the simplest and widely used methods where the mathematical model used for fitting is linear in the features and the target variable. The notion of dependency between the variables is easily captured by the covariance or correlation matrix which is easy to compute. Therefore, a popular measure for feature selection for linear regression is usually the squared multiple correlation (R^2-statistic) which can be computed in terms of the covariance matrices when the data is normalized (see Das and Kempe, 2008). Another widely used algorithm for feature selection in linear settings is Least Absolute Shrinkage and Selection Operator (LASSO) which uses l_1-norm based regularizer to promote sparsity in the selected features.

1.1.2 Diversity in the selected features

Apart from the measure that focuses on the dependency of the selected features and the target variable, another important aspect of feature selection is to capture diversity in the selected features in order to minimize redundancy. Diverse features often make more sense from interpreter's point of view as well. In addition, diversity in features often makes the model robust in presence of errors in the training data. Therefore, usage of diversity promoting regularizers in feature selection task is novel and it performs extremely well in many practical problems (see Das et al., 2012).

1.1.3 Usage of submodularity in feature selection

Submodularity is an elegant concept in mathematics that is relatively new but it enjoys the importance from many diverse fields that root in mathematical foundation. In Liu et al., 2013, the concept of submodularity was used in a feature selection algorithm that tries to select a subset of features that contains as much information as possible while being much smaller than the original set. Earlier in Das and Kempe, 2011 in the context of linear regression, it is shown that in feature selection task, while achieving exact submodularity using R^2 objective function is not possible, we can still get some optimality guarantee using the notion of *approximate submodularity.* Then in Das et al., 2012, a number of *submodular regularizers* were used along with the same objective function to promote diversity in the selected features. In that work, they had shown that while using such regularizers, we can still achieve some optimality guarantee for that objective function.

As for monotone submodular regularizers, Das et al., 2012 had used a simple Forward Selection algorithm, but the non-monotone case was rather complex. Since the goal for this particular setting is to maximize the objective function, the presence of non-monotone submodular regularizers makes the task hard. But in that work they had shown that using a local search algorithm similar to Feige et al., 2011 along with Forward Selection we can still reach an optimality guarantee. Even though the local search algorithm was originally designed for unconstrained maximization of non-monotone *submodular* objective, using a slight variation of that algorithm helped proving an optimality guarantee for their original objective function as well which is *not submodular* but *has a submodular regularizer* component. While the performance of this algorithm is extremely well, the complexity of this algorithm is relatively high due to the expensive local search algorithm.

In this present work, we explore the usage of a *linear time* algorithm for local search based on Buchbinder et al., 2012 which was designed for unconstrained submodular maximization (USM). It is interesting to note that, for using such an algorithm, the key idea is to *draw a relation* between *the solution set* returned by the original optimization algorithm (which works on a *submodular* objective) and *the optimal set* for the final objective function (whose optimal set can be different than the optimal set of the submodular component). For the local search based on Feige et al., 2011, this connection was explored in detail in the original paper. The main challenge in our approach was to obtain such a connection while using the linear time variant, which was not obvious from the description of the algorithm. The main contribution of this work is theorem 3.2.1, which forms the base of using a linear time algorithm for a generic objective function with submodular component. Following that, we show that we can still reach a provable optimality guarantee using this linear time variant, therefore reduce the overall complexity of the whole algorithm for diverse feature selection problem in linear regression setting.

1.2 Structure of the report

The content of this report is organized as follows. In chapter 2, we review the key mathematical concepts regarding submodularity, approximate submodularity and submodularity ratio. We also discuss the feature selection task for linear regression problem in detail. We review the squared multiple correlation or R^2-objective function and diversity promoting regularizers introduced in Das et al., 2012. In chapter 3, we discuss the local search algorithms for USM and their complexity and optimality guarantee. We discuss our theoretical results in this chapter as well present with proofs the optimality guarantee of our approach. Then in chapter 4, we discuss our experiments and results. Finally in chapter 5, we put concluding remarks for this work.

Chapter 2

Background and motivation

In this chapter, we start with a formal introduction to submodular functions in section 2.1. In subsection 2.1.2, we discuss the notion of submodularity ratio and approximate submodular functions. In the subsequent subsection 2.1.3, an algorithm and corresponding optimality result is reviewed for approximate submodular functions. Next, in section 2.2, we discuss the problem of feature selection in linear regression setup. In general we review that this is a hard problem but using squared multiple correlation as the objective function (in subsection 2.2.1) we can yet achieve some provable optimality guarantee. We also review some of the results from Das and Kempe, 2008 which establishes the robustness of this objective function to small perturbation in the training data. Finally in section 2.3, we discuss the importance of diversity in feature selection and why non-monotone submodular regularizers are strong candidates for such scenario. We also review the results from on optimality presented in Das et al., 2012 for approximately submodular objective functions with non-monotone diversity promoting regularizers.

2.1 Submodular functions

Submodular functions are *set functions* following certain properties. A *set function* is defined on a set X and it maps all possible $2^{|X|}$ subsets (usually written as 2^X for convenience) to a real value where the usual notion of $|.|$ denotes the cardinality of a set. There are a few equivalent definitions of submodularity. In the following we present the fundamental definition of submodularity (definition 2.1.1) and then in proposition 2.1.2 an alternative definition is presented which is more frequently used in the proofs of our relevant results.

Definition 2.1.1. *A set function $f : 2^X \to \mathbb{R}$ is submodular if and only if for all A, $B \subseteq X$,*

$$f(A) + f(B) \geq f(A \cup B) + f(A \cap B)$$

In almost all the cases, $f(\emptyset)$ is assumed to be 0. Therefore, for two disjoint sets $A, B \subseteq X$ the following proposition holds.

Proposition 2.1.1. *For all A, $B \subseteq X$ and $A \cap B = \emptyset$, $f(A) + f(B) \geq f(A \cup B)$*

Another interesting definition comes from the *property of diminishing returns* of submodular functions as promised by the following proposition. In Bach, 2011 the proof of its equivalency with definition 2.1.1 is provided. For the sake of completeness we present the proof here.

Proposition 2.1.2. *A set function $f : 2^X \to \mathbb{R}$ is submodular if and only if for all A, $B \subseteq X$ with $A \subseteq B$ and every $x \in X \backslash B$*

$$f(A \cup \{x\}) - f(A) \geq f(B \cup \{x\}) - f(B)$$

Proof. First we prove the forward direction of the claim. Let's consider $A \subseteq B$ and $x \in X \setminus B$. Representing $C = A \cup \{x\}$, we have $C \cup B = B \cup \{x\}$ and $C \cap B = A$. Therefore we can write

$$(f(A \cup \{x\}) - f(A)) - (f(B \cup \{x\}) - f(B)) = f(C) + f(B) - f(C \cup B) - f(C \cap B) \geq 0$$

To prove the reverse direction, we assume that proposition 2.1.2 holds. Let's consider n inequalities that can be written from this proposition in the form of $f(A \cup \{c_1, \cdots, c_k\}) - f(A \cup \{c_1, \cdots, c_{k-1}\}) \geq f(B \cup \{c_1, \cdots, c_k\}) - f(B \cup \{c_1, \cdots, c_{k-1}\})$ where $c_i \notin B$. Summing these up, for $C = \{c_1, \cdots c_n\}$ we have $f(A \cup C) - f(A) \geq f(B \cup C) - f(B)$. Finally, let's consider any $P, Q \subseteq X$. Taking $A = P \cap Q$ and $C = P \setminus Q$ and $B = Q$, we have $A \cup C = P$ and $B \cup C = P \cup Q$. Therefore $f(P) + f(Q) \geq f(P \cup Q) + f(P \cap Q)$. □

Throughout this report, we talk about *monotone* and *non-monotone* set functions. In the following we provide definitions for these.

Definition 2.1.2. *A set function $f : 2^X \to \mathbb{R}$ is called monotone if for all $A, B \subseteq X$ with $A \subseteq B$, we have $f(A) \leq f(B)$ or $f(A) \geq f(B)$.*

A set function is called *non-monotone* if it is not *monotone*.

2.1.1 Optimization of submodular functions

Since submodular functions arise mainly in discrete optimization problems, the aspect of minimization and maximization of submodular functions is one of the widely explored areas in theoretical development. For minimization, a convex relaxation of submodular set functions (known as Lovász extension) allows to find exact minimal solution in polynomial time (see Bach, 2011 for details). However finding the exact maxima of a submodular set functions has been proven to be NP-hard. Therefore a number of approximation algorithms exists in various settings (unconstrained, constrained) with some optimality guarantee. In an well known result from Nemhauser et al., 1978 it has been shown that for monotone submodular set functions under cardinality constraints a greedy forward selection algorithm can achieve a $(1 - \frac{1}{e})$ multiplicative approximation guarantee. For non-monotone case, in the unconstrained setting the best known result gives a $\frac{1}{3}$ approximation guarantee (see Feige et al., 2011 and Buchbinder et al., 2012) for non-negative submodular functions. In recent work by Buchbinder et al., 2014, an approximation algorithm exists that gives an approximation guarantee of $[\frac{1}{e} + 0.004, \frac{1}{2}]$ for maximizing non-monotone submodular functions with cardinality constraints. In chapter 3 we review local search based unconstrained submodular maximization algorithms.

2.1.2 Approximate submodular functions

In many practical scenario, the set function in hand may not be exactly submodular. In Das and Kempe, 2011 the notion of submodularity ratio was introduced for any non-negative set function $f : 2^X \to \mathbb{R}$.

Definition 2.1.3. *For a set X and some $k \geq 1$, the submodularity ratio $\gamma_{X,k}(f)$ is defined for any non-negative set function $f : 2^X \to \mathbb{R}$ as*

$$\gamma_{X,k}(f) = \min_{L \subseteq X, S : |S| \leq k, S \cap L = \emptyset} \frac{\sum_{x \in S} f(L \cup \{x\}) - f(L)}{f(L \cup S) - f(L)}$$

The submodularity ratio intuitively denotes how close a set function is to becoming submodular. It forms a necessary and sufficient condition for submodularity.

Lemma 2.1.1. *Any set function f is submodular if and only if $\gamma_{X,k}(f) \geq 1$*

We say that a set function f is *approximately submodular* if $\gamma_{X,k}(f) \to 1_{-}$. Proving an optimality guarantee in approximate submodular maximization is hard in general. However, for non-negative monotone approximate submodular functions defined on sets of real valued vectors, an optimality guarantee can be given in terms of its submodularity ratio. This type of functions is often very useful for feature selection problems.

2.1.3 Maximization of approximately submodular function

Let's consider a set of real valued vectors with cardinality n, $X = \{x_i\}_{i=1}^{n}$, where $x_i \in \mathbb{R}^N$ for some N. Let's assume that $f : 2^X \to \mathbb{R}$ is a non-negative monotone set function with submodularity ratio $\gamma_{X,k}$. Then a simple forward selection (also known as forward regression) algorithm outputs a solution set for the optimization problem

$$\arg\max_{S:|S| \leq k} f(S)$$

with provable optimality guarantee. This exact same algorithm achieves an optimality guarantee of $(1 - \frac{1}{e})$ for exact submodular set functions.

First we present the algorithm and then we present the theorem (based on Theorem 3.2 in Das and Kempe, 2011) showing its optimality guarantee.

FR algorithm

```
1: S_0 ← ∅
2: U ← {X_1, ··· , X_n}
3: for i ← 0 to i < k do
4:     X_j^* ← arg max_{X_j ∈ U \ S_i} f(S ∪ {X_j})
5:     S_{i+1} ← S_i ∪ {X_j^*}
6:     U ← U \ {X_j^*}
7: end for
return S_FR = S_k
```

The following theorem (based on Theorem 3.2 in Das and Kempe, 2011) shows the optimality guarantee. In the paper the proof is chalked out for a particular set function but it holds for generic non-negative monotone set functions defined on sets of real valued vectors as well.

Theorem 2.1.1. *The set S_{FR} selected by forward regression algorithm has an optimality guarantee of* $(1 - e^{-\gamma_{S_{FR},k}})$*, i.e.* $f(S_{FR}) \geq (1 - e^{-\gamma_{S_{FR},k}}) \cdot f(OPT)$ *where* $OPT \subseteq X$ *is the optimal set of cardinality* k*.*

The above result is very useful and presently best known result for subset selection with R^2 objective function which we discuss next.

2.2 Feature selection in linear regression

Given a set of features X, often selecting a subset $S \subseteq X$ with $|S| \leq |X|$ simplifies the learning and prediction task by reducing the problem size via removal of unnecessary or redundant features. This is a combinatorial optimization problem and in general is NP-hard (Amaldi and Kann, 1998). Although a number of filter methods exist which try to minimize a proxy error, yet no approximation guarantee can be provided while using the sum-squared error as an objective function.

In linear regression setting, we're given a set of N observations with regressors $X = \{\mathbf{x_i}\}_{i=1}^N$ with $\mathbf{x_i} \in \mathbb{R}^n$ and regressand $Z = \{z_i\}_{i=1}^N$ with $z_i \in \mathbb{R}$. The task is to fit a linear model and learn a predictor function $Z(X) = X\boldsymbol{\alpha} + \boldsymbol{\epsilon}$ ($\boldsymbol{\epsilon}$ represents error in observations). The optimal coefficient vector $\boldsymbol{\alpha}$ is usually obtained by the Least Squares method. For feature selection, the task is to therefore select a subset of the features which are expressed as $\varphi(x_j) = \{x_{i,j}\}_{i=1}^N$ where $x_{i,j}$ is the j^{th} entry in the feature vector $\mathbf{x_i}$. If the regressors X is represented as a $N \times n$ matrix, then the columns of X correspond to the features $\varphi(x_j)$. A feature selection algorithm then selects a subset of its columns, $S \in \mathbb{R}^{N \times k}$ ($k \leq n$), in order to fit the linear predictor $Z(S') = S\boldsymbol{\alpha} + \boldsymbol{\epsilon}$. In Das and Kempe, 2008 it is shown that using squared multiple correlation or R^2, a multiplicative approximation guarantee can be provided for the feature selection task in such setting. It is also established in the same work that this objective function is robust in presence of perturbation in training data which makes it a desired candidate for feature selection problem for linear regression.

2.2.1 R^2 objective function

The task here is to minimize mean-squared prediction error $\mathbb{E}[Z - Z'(S))^2]$ while selecting a subset $S \subseteq X$. This is equivalent to maximizing the squared multiple correlation $R^2 = \frac{\text{Var}(Z) - \mathbb{E}[(Z - Z'(S))^2]}{\text{Var}(Z)}$. This measure is widely used for a goodness-of-fit measure. Therefore the following objective function for feature selection is known to be a good candidate.

$$\arg\max_{S:|S| \leq k} R^2(S)$$

If we assume that the samples are normalized to have null mean and unit variance then $R^2 = 1 - \mathbb{E}[(Z - Z'(S))^2] = 1 - ||Z - Z'(S)||_2^2$. Let us consider the covariance matrix C with $C_{i,j} = \text{Cov}(X_i, X_j)$ and $\mathbf{b}$ with $b_j = \text{Cov}(Z, X_j)$. Let C_S and $\mathbf{b}_S$ be the sub-matrices with rows and columns corresponding to the set $S \subseteq X$ when the regressors and regressand have unit l_2-norm. It can be shown that $R^2 = \mathbf{b}_S(C_S)^{-1}\mathbf{b}_S$. In the following, we review some of the results provided in Das and Kempe, 2008 and Das and Kempe, 2011 which establishes the fact that R^2 is a good candidate for objective function in feature selection for linear regression.

Lemma 2.2.1. *Let's assume that the $k \times k$ sub-matrix of the covariance matrix, C_S is non-singular. Let κ denote the condition number for C_S where condition number of a matrix A is defined as $\kappa = ||A||_2||A^{-1}||_2$. If $E \in \mathbb{R}^{k\times k}$ is the error matrix with $\delta = \max_{i,j}(E_{i,j}) \leq \frac{1}{4\kappa k}$, then*

$$\frac{|\mathbf{b}_S^T C_S^{-1}\mathbf{b}_S - \mathbf{b}_S^T(C_S+E)^{-1}\mathbf{b}_S|}{|\mathbf{b}_S^T C_S^{-1}\mathbf{b}_S|} \leq \frac{4}{3}\kappa^2 k\delta$$

This result shows that as long as we have small perturbation in the covariance matrix, if the matrix is sufficiently well-conditioned, then the changes in R^2 objective function is small. This shows the robustness of the R^2 objective to perturbation in regressors.

R^2 is a set function which is monotone non-decreasing and non-negative. But it is not submodular in general. In Das and Kempe, 2008 it has been established that in absence of suppressor variables (which shadows the correlation of another variable with the regressand), R^2 becomes submodular, as stated by the following theorem.

Theorem 2.2.1. *In absence of suppressor variables in any $S \subseteq X$, R^2 objective function becomes submodular and hence one can achieve a multiplicative optimization guarantee of $(1-\frac{1}{e})$ via Forward Regression algorithm as proposed by Nemhauser et al., 1978.*

However, this condition is somewhat strict and therefore we can define a submodularity ratio for R^2 and search for approximate submodularity. The submodularity ratio for R^2 is defined as

$$\gamma_{X,k} = \min_{L\subseteq X, S:|S|\leq k, S\cap L=\emptyset} \frac{\sum_{x\in S} R^2_{L\cup\{x\}} - R^2_L}{R^2_{L\cup S} - R^2_L} = \min_{L\subseteq X, S:|S|\leq k, S\cap L=\emptyset} \frac{(\mathbf{b}_S^L)^T\mathbf{b}_S^L}{(\mathbf{b}_S^L)^T(C_S^L)^{-1}\mathbf{b}_S^L}$$

where C_S^L and $\mathbf{b}_S^L$ are covariance matrix and covariance vector for the corresponding entries in S and L. The following lemma from Das and Kempe, 2011 shows that the submodularity ratio is bounded below by the smallest eigenvalue in the covariance matrix.

Lemma 2.2.2. *Let's assume that the smallest eigenvalue of the covariance matrix C is $\lambda_{\min}$. Then $\gamma_{X,k} \geq \lambda_{\min}$.*

Since the covariance matrix is positive semi-definite, $\lambda_{\min}$ is always non-negative. In practice, the submodularity ratio for R^2 is close to 1 for sufficiently well-conditioned covariance matrices and therefore R^2 objective is claimed to be approximately submodular. Thus, by theorem 2.1.1, one can achieve, by simple forward selection algorithm, an optimality guarantee of $(1-e^{-\gamma_{S_{\mathrm{FR}},k}}) \geq (1-e^{-\lambda_{\min}})$.

2.3 Diversity promoting regularizers

Selecting the features as diverse as possible is often desired. First simple reason is that, it increases the interpretability of the selected features. Secondly, and most importantly, as Das et al., 2012 established, diverse features can make the task resistant to noise in the training data. The following objective function is used in Das et al., 2012 for diverse feature selection task.

$$\arg\max_{S:|S|\leq k} g(S) = R^2(S) + \nu f(S)$$

where $k > 0$ and $S \subseteq X$ and $\nu \geq 0$ is regularization constant and $f(S)$ is some diversity promoting set function. The term diversity is not uniquely defined and therefore they have used the following notion of diversity : For a fixed k

1. $f(S)$ is maximized when S is an orthogonal set of vectors (full rank)

2. $f(S)$ is minimized when the rank of the matrix S is the lowest

They have designed three such diversity promoting regularizers which are non-negative and submodular, out of which two are non-monotone in general. Because of the R^2 set function in the objective, the function $g(S)$ is not submodular. However, the following proposition shows that the submodularity ratio of $g(S)$ is bounded below by the submodularity ratio of R^2.

Lemma 2.3.1. *Let $f(S)$ be a non-negative submodular set function (monotone or non-monotone). Then for $g(S) = R^2(S) + f(S)$, the submodularity ratio*

$$\gamma_{X,k}(g) \geq \gamma_{X,k}(R^2) \geq \lambda_{\min}$$

where $\lambda_{\min}$ is the smallest eigenvalue of the covariance matrix C.

Next, we review the optimality guarantee for using such regularizers in the non-monotone case.

Theorem 2.3.1. *There exists a greedy local search algorithm for the objective function $g(S)$ with non-negative non-monotone submodular regularizers which obtains an approximation guarantee of $\frac{1-e^{-\frac{\gamma_{\hat{S},2k}(g)}{2}}}{2+(1-e^{-\frac{\gamma_{\hat{S},2k}(g)}{2}})(4+4\epsilon/n)}$ for some $\epsilon \geq 0$.*

In the next two section, we review the *non-monotone*, *submodular* regularizers used in Das et al., 2012.

2.3.1 Smoothed differential entropy regularizer

For some set S with $|S| \leq k$, the smoothed differential entropy regularizer is defined as

$$f_{\mathrm{de}}(S) = \sum_{i=1}^{|S|} \log_2(\delta + \lambda_i(C_S)) - 3k \log_2 \delta$$

where $\delta \geq 0$ is a smoothing constant and $\lambda_i(C_S)$ are the eigenvalues of the covariance sub-matrix C_S.

2.3.2 Spectral variance regularizer

For some set S with $|S| \leq k$, the spectral variance regularizer is defined as

$$f_{\text{sv}}(S) = 9k^2 - \sum_{i=1}^{|S|}(\lambda_i(C_S) - 1)^2$$

where $\lambda_i(C_S)$ are the eigenvalues of the covariance sub-matrix C_S as before.

None of the above two regularizers are monotone in general and yet they perform extremely well for selecting diverse features and show promising robustness to perturbation in training data (both the regressors and the regressand). In the greedy local search algorithm (GLS) mentioned above (see section 3.3 for the algorithm description), Das et al., 2012 used a comparatively expensive local search algorithm based on Feige et al., 2011 along with the forward regression algorithm in order to prove the optimality. A linear time local search exists for exact submodular objectives presented in Buchbinder et al., 2012. This serves for the motivation for finding optimality guarantee in using that algorithm for approximately submodular objective functions with non-monotone diversity promoting regularizers such as the ones used in Das et al., 2012.

Chapter 3

Local search for feature selection

The contents of this chapter are organized as follows. First, in section 3.1, we review the local search algorithm from Feige et al., 2011 for unconstrained submodular optimization (USM) for non-monotone submodular functions. We review some of the claims on optimality. In subsection 3.1.1, we review a slightly modified version of this algorithm that Das et al., 2012 used for maximization of the objective function $g(S) = R^2(S) + \nu f(S)$ for feature selection in linear regression setting as discussed in section 2.3. The optimality guarantee of this algorithm for such an objective function is reviewed. Next, we present the linear time local search algorithm from Buchbinder et al., 2012 and review some of the lemmas which helps us proving the optimality guarantee in section 3.2. Then in subsection 3.2.1 we present an algorithm based on this for maximization of set function $g(S)$. We prove the optimality guarantee in such approach using our proposed theorem 3.2.1. Finally, in section 3.3, we present the full greedy local search (GLS) framework presented in Das et al., 2012 for using this modified version of local search in the final algorithm. We also prove the optimality guarantee for the whole feature selection framework while using the linear time local search and compare with the guarantee presented in Das et al., 2012.

3.1 Unconstrained submodular maximization

Let's assume that we have a non-monotone non-negative submodular function $f : 2^X \to \mathbb{R}$. The unconstrained submodular optimization is then defined as below:

$$\hat{S} = \arg\max_S f(S)$$

In Feige et al., 2011, this has been claimed that finding an exact local maxima is hard in general (PLS-complete). Therefore, they have derived a local search algorithm which finds an approximate $(1+\alpha)$ local maxima. First we define the term and then we review the algorithm and discuss it's optimality guarantee and time complexity.

Definition 3.1.1. *Given a non-negative non-monotone submodular set function $f : 2^X \to \mathbb{R}$, a set S is called a $(1+\alpha)$-approximate local-optima, for some $\alpha \geq 0$, if*

$$(1+\alpha)f(S) \geq f(S \cup \{x\}), \forall x \in S$$

$$(1+\alpha)f(S) \geq f(S \setminus \{x\}), \forall x \notin S$$

Following from this definition, the next lemma (lemma 3.3 in Feige et al., 2011) turns out to be useful in proving the optimality guarantee.

Lemma 3.1.1. *If f is submodular set-function and S is a $(1+\alpha)$-approximate local-optima, then for any $T \subseteq S$ or $T \supseteq S$, then*

$$f(T) \leq (1+n\alpha)f(S)$$

Now we review the algorithm for USM for non-monotone non-negative submodular maximization.

LS algorithm for USM

- $S \leftarrow \arg\max_i f(X_i)$
- $U \leftarrow \{X_1, \cdots, X_n\}$
- **while** $\exists x \in U \setminus S$ s.t. $f(S \cup \{x\}) \geq (1+\epsilon/n^2)f(S)$ **do**
  - $S \leftarrow S \cup \{x\}$
- **end while**
- **while** $\exists x \in S$ s.t. $f(S \setminus \{x\}) \geq (1+\epsilon/n^2)f(S)$ **do**
  - $S \leftarrow S \setminus \{x\}$
- **end while**

return $S_{\text{LS}} = \arg\max_{T:T\in\{S, U\setminus S\}} f(T)$

It is obvious from the construction that at termination, this algorithm outputs a S_{LS} which is $(1+\epsilon/n^2)$ approximate local maxima. This algorithm gives an optimality guarantee of $(\frac{1}{3} - \frac{\epsilon}{n})$ and takes $\mathcal{O}(\frac{1}{\epsilon}n^3 \log n)$. These proofs are not obvious and hence we provide them from Feige et al., 2011 for the sake of completeness.

Theorem 3.1.1. *If f is non-negative, submodular, then for* any *set $T \subseteq U$ and any $\epsilon > 0$, the LS algorithm forms a set S, s.t. $(2+2\epsilon/n)f(S) + f(U \setminus S) \geq f(T)$.*

Proof. Let's consider a set $T \subseteq U$ and let $\alpha = \epsilon/n^2$. For the set $S \cup T$, we can write $f(S \cup T) \leq (1+n\alpha)f(S)$ since $S \cup T \supseteq S$ (from lemma 3.1.1). Similarly, $f(S \cap T) \leq (1+n\alpha)f(S)$ since $S \cap T \subseteq S$. Also, from submodularity,

$$\begin{aligned}
f(S \cup T) + f(U \setminus S) &\geq f(S \cup T \cup (U \setminus S)) + f((S \cup T) \cap (U \setminus S)) && (3.1)\\
&\geq f(U) + f(T \setminus S) && (3.2)\\
&\geq f(T \setminus S) && (3.3)
\end{aligned}$$

The last line follows from the non-negativity of f. Similarly,

$$\begin{aligned}
f(S \cap T) + f(T \setminus S) &\geq f((S \cap T) \cup (T \setminus S)) + f((S \cap T \cap (U \setminus S)) && (3.4)\\
&\geq f(T) + f(\emptyset) && (3.5)\\
&\geq f(T) && (3.6)
\end{aligned}$$

Therefore,

$$\begin{aligned}
2(1+n\alpha)f(S) + f(U \setminus S) &\geq f(S \cap T) + f(S \cup T) + f(U \setminus S) && (3.7)\\
&\geq f(S \cap T) + f(T \setminus S) && (3.8)\\
&\geq f(T)
\end{aligned}$$

□

Theorem 3.1.2. *If f is non-negative, submodular, then the LS algorithm gives an optimality guarantee of* $(\frac{1}{3} - \frac{\epsilon}{n})$

Proof. From theorem 3.1.1, we can write, for the optimal set $C \subseteq U$, $(2 + 2\epsilon/n)f(S) + f(U \setminus S) \geq f(C)$. Let's assume that the set returned by the algorithm is S which is when $f(S) \geq f(U \setminus S)$. Therefore, for this case $(2+2\epsilon/n)f(S)+f(S) \geq (2 + 2\epsilon/n)f(S) + f(U \setminus S) \geq f(C)$, and therefore $f(S) \geq \frac{1}{(3+2\epsilon/n)} f(C)$ which is at least $(\frac{1}{3} - \frac{\epsilon}{n})f(C)$ for all $\epsilon \geq 0$. The proof for $U \setminus S$ follows similarly. □

Theorem 3.1.3. *The LS algorithm described has complexity* $\mathcal{O}(\frac{1}{\epsilon}n^3 \log n)$.

Proof. Let's assume that $f(\{\hat{x}\}) = \max_i f(\{x_i\})$, *i.e.* it is the maximum of all singletons. Since the function f is submodular, its submodularity ratio $\gamma \geq 1$. Therefore,

$$\min_{L \subseteq X, S: |S| \leq k, S \cap L = \emptyset} \frac{\sum_{x \in S} f(L \cup \{x\}) - f(L)}{f(L \cup S) - f(L)} \geq 1$$

Without any loss of generality, we take $L = \emptyset$ and $S = C$, *i.e.* the optimal set where $k = |C|$. Thus we have

$$\frac{\sum_{x \in C} f(\{x\}) - f(\emptyset)}{f(C) - f(\emptyset)} \geq 1$$

which leads to $nf(\{\hat{x}\}) \geq |C|f(\{\hat{x}\}) \geq \sum_{x \in C} f(\{x\}) \geq f(C)$. Therefore, the solution set returned by the algorithm S must follow $nf(\{\hat{x}\}) \geq f(S)$.

Now, let's assume that the algorithm ran up to k iterations and in each step the function value is increased by up to a multiplicative factor of at least $(1 + \epsilon/n^2)$. Therefore, for the final set, S, $f(S) \geq (1 + \epsilon/n^2)^k f(\{\hat{x}\})$ where $\hat{x}$ was selected in the initial arg max computation over all the singletons. Thus $nf(\{\hat{x}\}) \geq (1 + \epsilon/n^2)^k f(\{\hat{x}\})$. Therefore, $k = \mathcal{O}(\frac{1}{\epsilon}n^2 \log n)$ and since therefore the overall complexity is $\mathcal{O}(\frac{1}{\epsilon}n^3 \log n)$. □

3.1.1 Unconstrained approximate submodular maximization

In Das et al., 2012, an algorithm which is very similar to the one described in previous section was used to find an approximate maxima for the objective function $g(S) = R^2(S) + \nu f(S)$.

LS algorithm for maximizing $g(S)$.

1: $S \leftarrow \arg\max_i f(X_i)$
2: $U \leftarrow \{X_1, \cdots, X_n\}$
3: **while** $\exists x \in U \setminus S$ s.t. $f(S \cup \{x\}) \geq (1 + \epsilon/n^2)f(S)$ **do**
4: $\quad S \leftarrow S \cup \{x\}$
5: **end while**
6: **while** $\exists x \in S$ s.t. $f(S \setminus \{x\}) \geq (1 + \epsilon/n^2)f(S)$ **do**
7: $\quad S \leftarrow S \setminus \{x\}$
8: **end while**
return $S_{\text{LS}} = \arg\max_{T: T \in \{S, U \setminus S, U\}} g(T)$

It is obvious that this algorithm has the same asymptotic run-time as of the previous LS algorithm for USM. We now review the optimality guarantee in this by the following theorem in Das et al., 2012 (theorem 7). Again, for the sake of completeness we present the proof of this theorem.

Theorem 3.1.4. *The above LS algorithm gives a* $\frac{1}{4+\frac{4\epsilon}{n}}$ *multiplicative approximation guarantee for* $\arg\max_S g(S)$.

Proof. Let's assume that the optimal set for $g(S)$ is C. When the algorithm terminates, we therefore have a set S obtained by the LS algorithm above. Now, from theorem 3.1.1, we can write that $(2+2\epsilon/n)f(S)+f(U\setminus S)\geq f(C)$. Since $g(S)=R^2+\nu f(S)$ and R^2 is non-negative and $\nu\geq 0$, we have $\nu(2+2\epsilon/n)f(S)+(2+2\epsilon/n)R^2+\nu f(U\setminus S)+R^2(U\setminus S)\geq \nu f(C)$ or $(2+2\epsilon/n)g(S)+g(U\setminus S)\geq \nu f(C)$. Now, since R^2 is monotone as well, we have $R^2(U)\geq R^2(C)$ since $C\subseteq U$. Thus, $(2+2\epsilon/n)g(S)+g(U\setminus S)+R^2(U)\geq \nu f(C)+R^2(C)=g(C)$. Finally, we have f which is non-negative, and therefore, $(2+2\epsilon/n)g(S)+g(U\setminus S)+R^2(U)+\nu f(U)\geq g(C)$ or

$$(2+2\epsilon/n)g(S)+g(U\setminus S)+g(U)\geq g(C)$$

Then, let's assume that S was selected in the final arg max computation. Therefore $g(S)\geq g(U\setminus S)$ and $g(S)\geq g(U)$. So, $(2+2\epsilon/n)g(S)+g(S)+g(S)\geq(2+2\epsilon/n)g(S)+g(U\setminus S)+g(U)\geq g(C)$ or $g(S)\geq\frac{1}{(4+2\epsilon/n)}g(C)\geq\frac{1}{(4+4\epsilon/n)}g(C)$. Similar argument follows for $U\setminus S$ and U. □

3.2 Linear time unconstrained submodular maximization

In Buchbinder et al., 2012, a linear time deterministic algorithm was proposed providing an approximation guarantee $\frac{1}{3}.f(OPT)$ for USM where the objective is non-negative non-monotone. First, let us discuss briefly how the linear-time solution works.

Linear-time LS algorithm for maximizing $f(S)$

$X_0 \leftarrow \emptyset$
$Y_0 \leftarrow \{X_1,\cdots,X_n\}$
for $i\leftarrow 1$ to $i\leq|U|$ **do**
  $a_i\leftarrow f(X_{i-1}\cup\{x_i\})-f(X_{i-1})$
  $b_i\leftarrow f(Y_{i-1}\setminus\{x_i\})-f(Y_{i-1})$
  if $a_i\geq b_i$ **then**
    $X_i\leftarrow X_{i-1}\cup\{x_i\}$
    $Y_i\leftarrow Y_{i-1}$
  else
    $X_i\leftarrow X_{i-1}$
    $Y_i\leftarrow Y_{i-1}\setminus\{x_i\}$
  end if
end for
return X_n or Y_n (Both are same)

The approximate optimality guarantee proof follows from two lemmas presented in Buchbinder et al., 2012:

Lemma 3.2.1. *For every $i \in [1, n]$, $a_i + b_i \geq 0$.*

Lemma 3.2.2. *For every $i \in [1, n]$,*

$$f(OPT_{i-1}) - f(OPT_i) \leq f(X_i) - f(X_{i-1}) + f(Y_i) - f(Y_{i-1})$$

where OPT_i is defined as $(OPT \cup X_i) \cap Y_i$. Thus, $OPT_0 = OPT$ and $OPT_n = X_n = Y_n$.

The final proof then works as summing up $\forall i \in [1, n]$, the inequality from this lemma.

$$\begin{aligned} f(OPT_0) - f(OPT_n) &\leq f(X_n) - f(X_0) + f(Y_n) - f(Y_0) &(3.9)\\ &\geq f(X_n) + f(Y_n) &(3.10)\end{aligned}$$

and therefore, $f(S) = f(X_n) = f(Y_n) = f(OPT)/3$.

However the relation between the solution set returned by the algorithm S while solving f, with any other set $T \subseteq U$ is not obvious from the construction of this algorithm. But, to use a linear time local search for finding $\arg\max_S g(S)$, we need to establish such a relation since optimal set of f and g are different. Therefore, we propose the following theorem establishing the base of our proofs and motivation towards the proposed algorithm.

Theorem 3.2.1. *The solution set returned by the USM linear time local search algorithm, S, is a $(1+\alpha)$-approximate local optima where $\alpha = \max_i(\frac{1}{f(S)}(a_i+b_i)) \in [0, 2]$ where $a_i = f(X_{i-1} \cup \{x_i\}) - f(X_{i-1})$ and $b_i = f(Y_{i-1} \setminus \{x_i\}) - f(Y_{i-1}))$*

Proof. First, we notice that for any $i \in [1, n]$, the final set selected by the algorithm $S = X_n = Y_n$ would satisfy $X_i \subseteq S \subseteq Y_i$.

For $T \subseteq S$, we can think of a chain $T = T_1 \subseteq T_2 \cdots \subseteq T_k = S$, where $T_j \setminus T_{j-1} = \{x_j\}$. Now for each such element $x_j \notin T_{j-1}$ and $x_j \in S$, we can write, from the property of diminishing returns of the submodular function f (proposition 2.1.2),

$$f(T_j) - f(T_{j-1}) \geq f(S) - f(S \setminus \{x_j\}) \geq f(Y_{j-1}) - f(Y_{j-1} \setminus \{x_j\}) = -b_j$$

Summing up, we get, $f(S) - f(T) \geq -\sum_j b_j$ or

$$f(T) \leq f(S) + \sum_j b_j$$

Notice that these elements have been added in the final solution set, S, by the algorithm. Therefore $\forall x_j \in S$, $a_j \geq b_j$. So we have

$$f(T) \leq f(S) + \sum_j b_j \leq f(S) + \sum_j a_j \leq f(S) + \sum_j (a_j + b_j)$$

Now let's consider the simplest case where $T = S \setminus \{x_k\}$ for some k. So we have $f(T) = f(S \setminus \{x_k\}) \leq f(S) + a_k + b_k$. Since $\forall i$, $a_i + b_i \geq 0$, we can write this as

$$f(S \setminus \{x_k\}) \leq (1 + \alpha_k)f(S)$$

for $\alpha_k = \frac{1}{f(S)}(a_k + b_k) \geq 0$.

Similarly, for $T \supseteq S$, we can consider the chain $S = T_1 \subseteq T_2 \cdots \subseteq T_m = T$, where $T_j \setminus T_{j-1} = \{x_j\}$ and we find that

$$f(T_j) - f(T_{j-1}) \leq f(S \cup \{x_j\}) - f(S) \leq f(X_{j-1} \cup \{x_j\}) - f(X_{j-1}) = a_j$$

and therefore

$$f(T) \leq f(S) + \sum_j a_j$$

Now $\forall x_j \notin S$, we have, from the algorithm, $a_i < b_i$. So, going by similar logic we can find, for some $T = S \cup \{x_k\}$,

$$f(S \cup \{x_k\}) \leq (1 + \alpha_k) f(S)$$

for $\alpha_k = \frac{1}{f(S)}(a_k + b_k) \geq 0$. Choosing $\alpha = \max_i(\alpha_i) = \max_i(\frac{1}{f(S)}(a_i + b_i)) \geq 0$, we can write,

- for any $x_k \in S$, $f(S \setminus \{x_k\}) \leq (1 + \alpha) f(S)$
- and for any $x_k \notin S$, $f(S \cup \{x_k\}) \leq (1 + \alpha) f(S)$.

Therefore S is a $(1 + \alpha)$-approximate local optima according to definition 3.1.1. Now it is left to prove the upper bound of α. To prove that, we notice that the the algorithm always increases the function value of f in each iteration. This is easy to verify, since if it didn't then at least at one step in the algorithm we'd have got the gains, a_i and b_i both negative but according to lemma 3.2.1, $a_i + b_i \geq 0$. Thus $f(S) \geq f(X_i)$ and $f(S) \geq f(Y_i)$. Therefore, $\max_i \frac{1}{f(S)}(f(X_{i-1} \cup \{x_i\}) - f(X_{i-1}) + f(Y_{i-1} \setminus \{x_i\}) - f(Y_{i-1})) \leq 2$. □

3.2.1 Proposed algorithm for approximate submodular maximization

For our objective function of interest, $g(S) = R^2 + \nu f(S)$, we propose the following algorithm which is very similar to the one presented in Buchbinder et al., 2012 for USM with non-negative, non-monotone submodular objective. First we present the algorithm and then we analyze the optimality guarantee.

Theorem 3.2.2. *The proposed LS algorithm gives a* $\frac{1}{4+2n\alpha} \in [\frac{1}{4}, \frac{1}{4+4n}]$ *multiplicative approximation guarantee for* $\arg\max_S g(S)$ *where* $\alpha = \max_i(\frac{1}{f(S)}(a_i + b_i)) \in [0, 2]$ *where* $a_i = f(X_{i-1} \cup \{x_i\}) - f(X_{i-1})$ *and* $b_i = f(Y_{i-1} \setminus \{x_i\}) - f(Y_{i-1}))$ *in the above algorithm.*

Proof. First, we notice that using lemma 3.1.1 and theorem 3.2.1, we can write, for any $T \subseteq S$ or $T \supseteq S$, then

$$f(T) \leq (1 + n\alpha) f(S) \leq (1 + 2n) f(S)$$

Therefore, from lemma 3.1.1, we can write for *any* set $T \subseteq U$

$$(2 + 4n) f(S) + f(U \setminus S) \geq (2 + 2n\alpha) f(S) + f(U \setminus S) \geq f(T)$$

Linear-time LS algorithm for maximizing $g(S)$

```
X_0 ← ∅
Y_0 ← {X_1, ··· , X_n}
for i ← 1 to i ≤ |U| do
    a_i ← f(X_{i−1} ∪ {x_i}) − f(X_{i−1})
    b_i ← f(Y_{i−1} \ {x_i}) − f(Y_{i−1})
    if a_i ≥ b_i then
        X_i ← X_{i−1} ∪ {x_i}
        Y_i ← Y_{i−1}
    else
        X_i ← X_{i−1}
        Y_i ← Y_{i−1} \ {x_i}
    end if
end for
return S_LS = arg max_{T:T∈{X_n,U\X_n,U}} g(T)
```

Thus, following the proof in theorem 3.1.4, we can write that

$$\max(g(S), g(U \setminus S), g(U)) \geq \frac{1}{4 + 2n\alpha} g(C) \geq \frac{1}{4 + 4n} g(C)$$

□

In practice, the value of α is $\ll 2$ and also this guarantee does not depend on the ϵ parameter in the local search algorithm. In section 4.1.1, we show a comparative plot of the these two algorithms on a synthetic data set.

3.3 Greedy local search algorithm for feature selection

In this section we review the greedy local search algorithm framework proposed by Das et al., 2012 where the FR algorithm (see subsection 2.1.3) is used along with LS. First, we present the algorithm and review a theorem from the same proving the optimality guarantee for this algorithm. Then we propose a theorem proving the optimality guarantee if we replace the local search with the proposed local search in subsection 3.2.1.

GLS algorithm for maximizing $g(S)$

```
U ← {X_1, ··· , X_n}
S_1 ← FR(U)
S'_1 ← LS(S_1)
S_2 ← FR(U \ S_1)
return S_GLS = arg max_{T:T∈{S_1,S'_1,S_2}} g(T)
```

First, we mention the following two results (theorem 5 and lemma 8) from Das et al., 2012 which helps in proving the optimality bound.

Theorem 3.3.1. *Let $T \subseteq U$ with $|T| \leq k$. Then the set S selected by the FR algorithm satisfies*

$$g(S) \geq (1 - e^{-\frac{\gamma_{S,2k}}{2}})g(S \cup T)$$

Lemma 3.3.1. *Let's assume that we have sets $C, S_1 \subseteq U$. Also, let's assume that $C' = C \setminus S_1$ and $S_2 \subseteq U \setminus S_1$. Then $g(S_1 \cup C) + g(S_2 \cup C') + g(S_1 \cap C) \geq g(C)$.*

Finally we reproduce the theorem from Das et al., 2012 showing the optimality guarantee which works for both the local search algorithms discussed above.

Theorem 3.3.2. *The GLS algorithm outputs a solution set S with optimality guarantee*

$$\frac{1 - e^{-\frac{\gamma_{S,2k}}{2}}}{2 + (1 - e^{-\frac{\gamma_{S,2k}}{2}})(4 + 2n\alpha)} \geq \frac{1 - e^{-\frac{\gamma_{S,2k}}{2}}}{6 + 2n\alpha}$$

Proof. Let $C \subseteq U$, $|C| \leq k$ is the optimal solution for $\arg\max_{S:|S|\leq k} g(S) = R^2_Z(S) + \nu f(S)$. Applying theorem 3.3.1, we obtain that $g(S_1) \geq \lambda g(S_1 \cup C)$, where $\lambda = (1 - e^{-\frac{\gamma_{S,2k}}{2}})$. Now we consider two cases.
Case 1: $g(S_1 \cap C) \geq \delta g(C)$ for some $0 \leq \delta \leq 1$. In this case, the local search algorithm selects a set S'_1 with optimality guarantee $f(S'_1) \geq \frac{\delta}{4+2n\alpha} g(C)$ following theorem 3.1.4 and 3.2.2.
Case 2: $g(S_1 \cap C) < \delta g(C)$. Therefore, $\lambda g(S_1 \cap C) - \lambda\delta g(C) < 0$. Therefore, adding this to the RHS of $g(S_1) \geq \lambda g(S_1 \cup C)$, we get $g(S_1) \geq \lambda g(S_1 \cup C) + \lambda g(S_1 \cap C) - \lambda\delta g(C)$. Also, we have, applying 3.3.1 for the second FR, $g(S_2) \geq \lambda g(S_2 \cup (C \setminus S_1))$. Adding these two equations, we get $g(S_1) + g(S_2) \geq \lambda g(S_1 \cup C) + \lambda g(S_1 \cap C) + \lambda g(S_2 \cup (C \setminus S_1)) - \lambda\delta g(C)$. Therefore, applying lemma 3.3.1, we get $g(S_1) + g(S_2) \geq \lambda g(C) - \lambda\delta g(C) = \lambda(1-\delta)g(C)$. This shows that $\max(g(S_1), g(S_2)) \geq \frac{\lambda(1-\delta)}{2} g(C)$. Now combining the two cases, we have an optimality guarantee

$$\max(g(S_1), g(S'_1), g(S_2)) = \max\{\frac{\delta}{4 + 2n\alpha}, \frac{\lambda(1-\delta)}{2}\}g(C)$$

Setting $\delta = \frac{\lambda(4+2n\alpha)}{2+\lambda(4+2n\alpha)}$ we achieve the guarantee of $\frac{1-e^{-\frac{\gamma_{S,2k}}{2}}}{2+(1-e^{-\frac{\gamma_{S,2k}}{2}})(4+2n\alpha)}$ for both these cases. which is at least $\frac{1-e^{-\frac{\gamma_{S,2k}}{2}}}{6+2n\alpha}$ since $1 - e^{-\frac{\gamma_{S,2k}}{2}} \leq 1$. □

Setting $\alpha = 1/2n$, we obtain the optimality guarantee promised in Das et al., 2012 when using the local search version based on Feige et al., 2011. When using linear time local search based on Buchbinder et al., 2012, we therefore obtain a guarantee $\left[\frac{1-e^{-\frac{\gamma_{S,2k}}{2}}}{6}, \frac{1-e^{-\frac{\gamma_{S,2k}}{2}}}{6+4n}\right]$.

Chapter 4

Experiments and results

We describe our experimental setups and results in this chapter.

4.1 Experimental setups

We ran the experiment on several standard data sets including

- `housing` data set
  - (`https://archive.ics.uci.edu/ml/datasets/Housing`)
  - comprises of 13 attributes and 506 instances
- `wine` data set
  - (`https://archive.ics.uci.edu/ml/datasets/Wine+Quality`)
  - comprises of 11 attributes and 4898 instances
- `indoor` data set
  - (`https://archive.ics.uci.edu/ml/datasets/UJIIndoorLoc`)
  - comprises of 528 attributes and 21048 instances
- `blog` data set
  - (`https://archive.ics.uci.edu/ml/datasets/BlogFeedback`)
  - comprises of 280 attributes and 60021 instances
- `mnist` data set that is used in Das et al., 2012 in the same fashion in order to be able to compare our results with theirs
  - (`http://yann.lecun.com/exdb/mnist/`)
  - 1000 $\sim$ 10000 randomly selected instances out of 60000 with 784 features

Our data (both regressors and regressand) are column-wise normalized to have unit l_2-norm. We also performed an experiment on a synthetic data set with just the local search algorithms used in Das et al., 2012 and our proposed algorithm for the approximately submodular objective function $g(S)$. The synthetic data set was obtained by randomly generating a matrix with entries i.i.d. samples from $U(0, 1)$. We randomly added some zero columns into this data set in order to make the functions behave like non-monotone in the input set. This matrix was then column-wise normalized as well. The experiments were performed on a Lenovo Thinkpad T430 Intel® Core™ i5-3230M CPU @ 2.60GHz with 8GB DDR3 @ 1600MHz running Fedora Linux 3.17.6-300.fc21.x86_64).

4.1.1 Results

As we can see from figure 4.1, the linear time LS algorithm for $g(S)$ shows similar performance except for a few cases with the LS algorithm used in Das et al., 2012 on the synthetic data set in terms of the size of the set it returns and in terms of the function value on the output set, $g(\hat{S})$ but takes significantly less time as the size of the input increases. In the figure 4.2, we see the whole feature selection framework in action on `mnist` data set with 10000 instances selected uniformly randomly from total 60000 instances with smoothed differential entropy regularizer with $\nu = 0.0001$, $\delta = 0.1$. We have varied the number of target features from 10 to 100. We then fit a linear model using the selected features and measured test-statistic on the test data. Here we can observe that the R^2 statistic ≥ 0.8 even with just 10 features. This performance is similar to the one proposed by Das et al., 2012.

4.1.2 Tesseract - An open source library

As a course of development, we built an open source library under MIT license, Tesseract, which is hosted in Github (`https://github.com/lambday/tesseract`). The implementation is done in C++ for speed and efficient memory management. We have used Eigen3 (`http://eigen.tuxfamily.org/`), another open source library, for the linear algebra operations in the algorithms. The features of this library are

- showcasing the different algorithms discussed in this report
- a unit testing suite
- an integration testing suite
- memcheck suite for all the unit/integration tests
- different measures of performance on test-data including
  - sum of squared errors
  - Pearson's correlation
  - squared multiple correlation measure or R^2 statistic
- doxygen generated documentation and reference manual

Since for the algorithms we implemented here, all the computations are performed on the covariance matrix of the normalized regressors, $X \in \mathbb{R}^{N \times n}$, and regressand, $Z \in \mathbb{R}^N$, we compute and store the the covariance matrix $C \in R^{(n+1)\times(n+1)}$ and

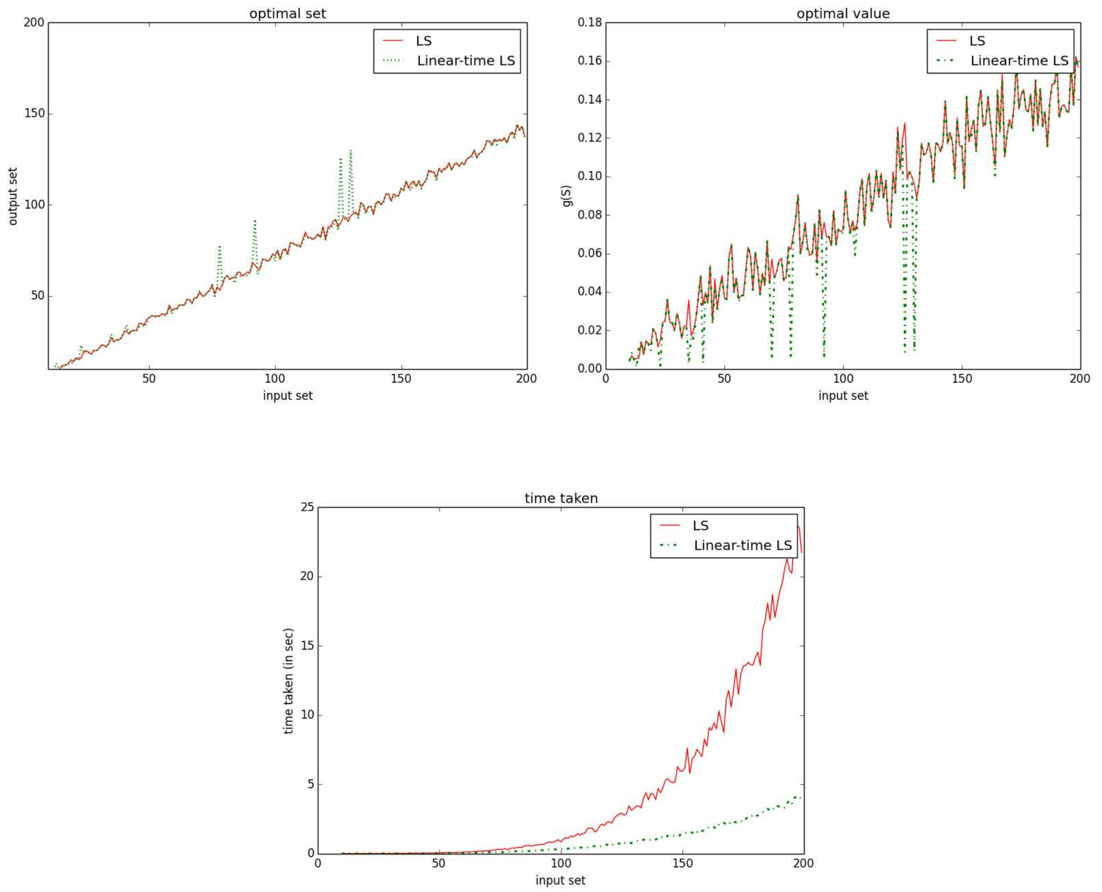

Figure 4.1: Comparison of performance of LS algorithms on $g(S)$ with smoothed differential entropy regularizer with $\nu = 0.0001$, $\delta = 0.1$. (a) This shows the size of the output set that the LS algorithms return (b) Plot of $g(\hat{S})$ where $\hat{S}$ is the output of the LS algorithms (c) Time taken by the LS algorithms

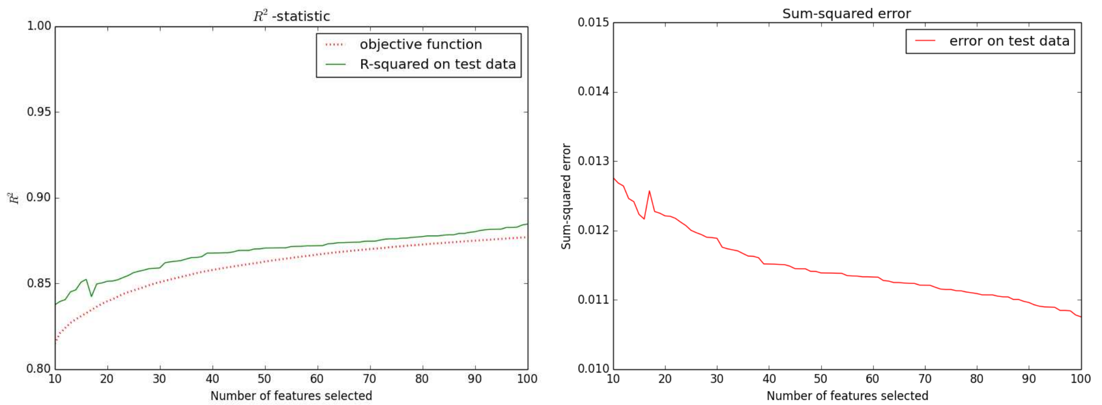

Figure 4.2: GLS algorithm with linear LS with smoothed differential entropy regularizer with $\nu = 0.0001$, $\delta = 0.1$. (a) This shows R^2-statistic on test data (b) This shows the sum-squared error on test data

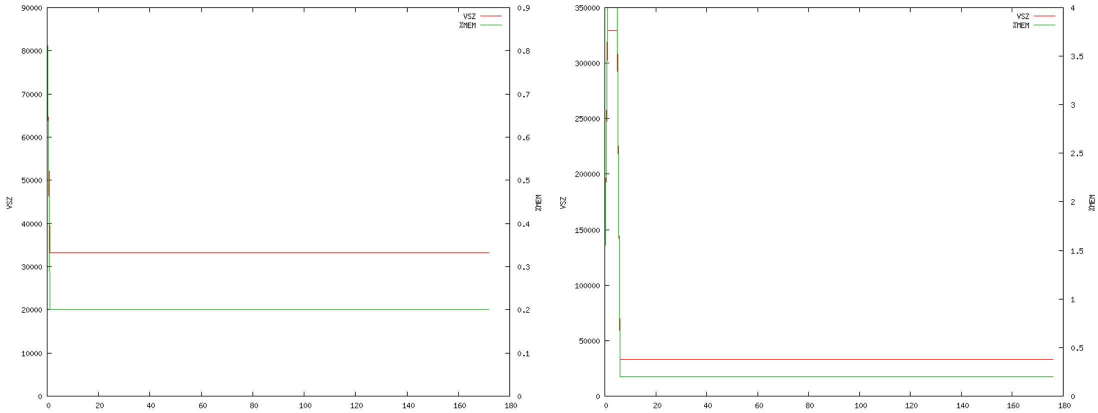

Figure 4.3: Memory usage by the Tesseract on `mnist` data set (784 features) with target features 20 (a) $10,000$ examples (b) $50,000$ examples

discard the data. Since usually $n \ll N$, This reduces the overall memory usage of the library and also makes it virtually independent on the number of examples used, N. In figure 4.3 we show the memory usage by the library running on `mnist` data set for different number of examples. As seen from this plots, the memory usage is constant after the initial burst where we compute the covariance matrix and is practically independent on the number of examples.

Chapter 5

Conclusion and remarks

We have discussed in detail the problem of diverse feature selection in linear regression setting. We have established a theoretical result which shows that in a greedy local search (GLS) framework proposed by Das et al., 2012, using a linear time local search variant based on Buchbinder et al., 2012 instead of a rather expensive variant similar to Feige et al., 2011, we can still reach provable optimality guarantee. Unlike the optimality guarantee in Das et al., 2012, our optimality guarantee does not depend on external parameters to the algorithm.

Also, the present work allows to increase scalability of the entire GLS framework if we use a parallel version of the linear time local search algorithm proposed by Pan et al., 2014. This was not possible before due to the sequential nature of the algorithm used in the original paper by Das et al., 2012. This parallel version selects in each step the exact same element as the one proposed in Buchbinder et al., 2012 and therefore the optimality guarantee proposed in the present work holds readily in that case. It would be interesting to see how the usage of the parallel version scales up the size of the problems the framework can handle.

However, from our experiments we have also noticed that the regularizer being non-monotone is a specific case which is hard to observe in many standard data sets. In many cases, although theoretically non-monotone, the regularizers used in our experiments behaved like monotone non-decreasing functions. Therefore, in practice, a standard Forward Regression algorithm suffices to achieve feature subset with sufficiently low error rate. Also, extreme care has to be taken for the regularization constant ν.

Another important thing to note is that in the complexity analysis of the local search algorithms, it is assumed that the function calls are oracle calls in $\mathcal{O}(1)$ which is not the case in practice. Therefore, additional complexity gets added while applying these algorithms in real life specially for larger sized inputs.

To conclude, the usage of submodularity in the feature selection problem makes it interesting from theoretical point of view. It would be interesting to see whether we can design such approach for feature selection in more complex scenario. In particular, in kernel-based feature selection algorithms, such as BAHSIC (see Song et al., 2007), which uses Backward Elimination approach to select target features in kernel-mapped space using Hilbert-Schmidt Independence Criterion (HSIC) (see Gretton et al., 2005) as dependency measure, it would be interesting to see if we can achieve some optimality guarantee using submodularity.

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